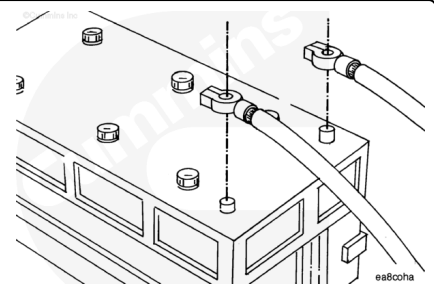


Trainings ▾

Remove

⚠ WARNING ⚠

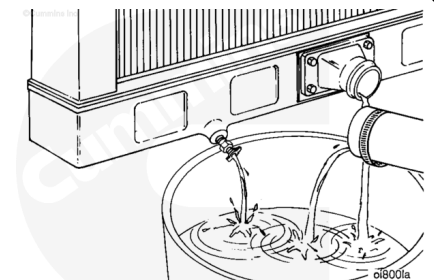
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Disconnect the batteries and air supply line to the air starter to prevent accidental starting. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/57/57-013-009.html>).

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

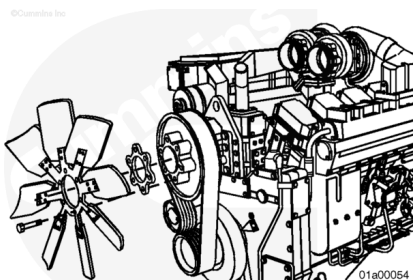


⚠ WARNING ⚠

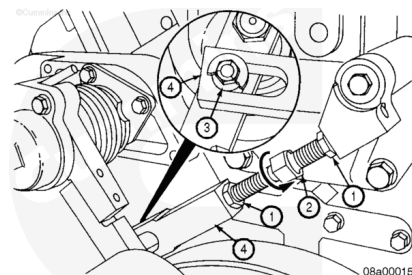
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/57/57-008-018.html>) in Section 8.

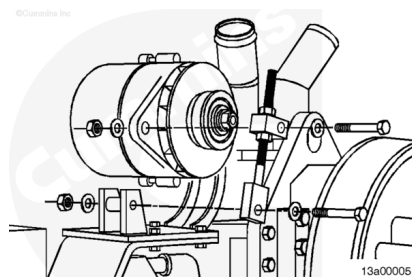
Remove the fan and fan spacer(s). Refer to Procedure 008-040 in Section 8 (</qs3/pubsys2/xml/en/procedures/57/57-008-040.html>).



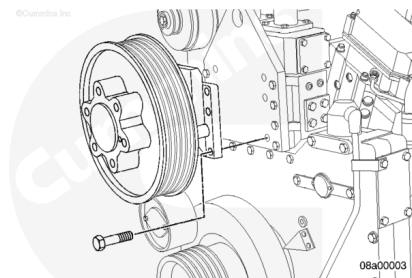
Remove the fan belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/57/57-008-002-tr.html) in Section 8.



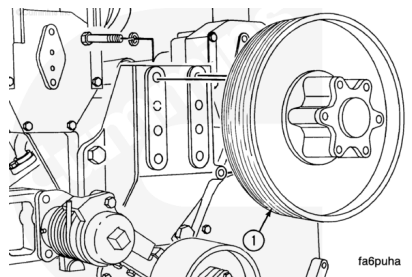
Remove the alternator belt and alternator. Refer to Procedure 013-001 in Section 13 (/qs3/pubsys2/xml/en/procedures/57/57-013-001-tr.html).



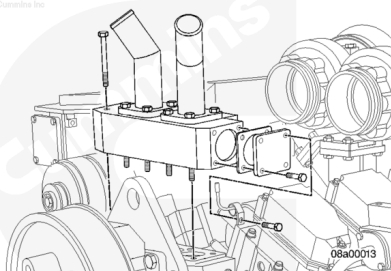
Remove the fan hub. Refer to Procedure 008-036 (/qs3/pubsys2/xml/en/procedures/57/57-008-036-tr.html) in Section 8.



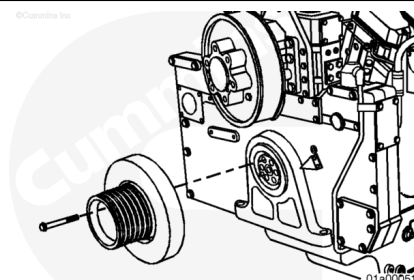
Remove the fan hub idler.



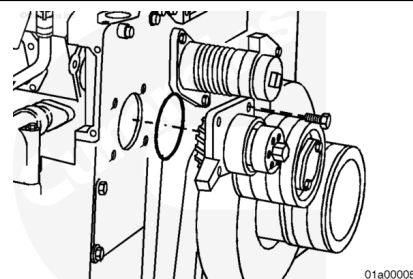
Remove the thermostat housing. Refer to Procedure 008-015 in Section 8 (/qs3/pubsys2/xml/en/procedures/57/57-008-015.html).



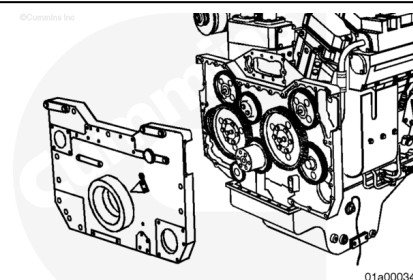
Remove the vibration damper. Refer to Procedure 001-052 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-052.html>).



Remove the charging system alternator drive. Refer to Procedure 009-008 in Section 9. (</qs3/pubsys2/xml/en/procedures/57/57-009-008.html>)



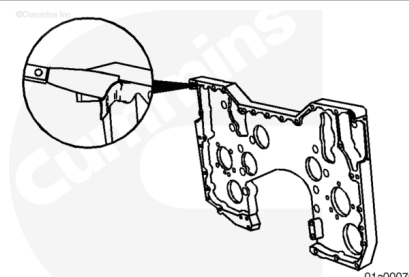
Remove the gear cover. Remove and discard the gasket. The gear cover has 23 capscrews in the front of the cover.



Clean

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Remove the RTV sealant and gasket from the lower and rear surfaces of the gear cover.

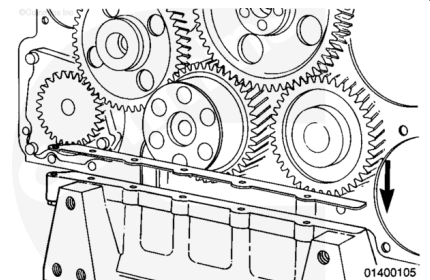
Use a steam cleaner or solvent tank to clean the gear cover.

Dry with compressed air.

Install

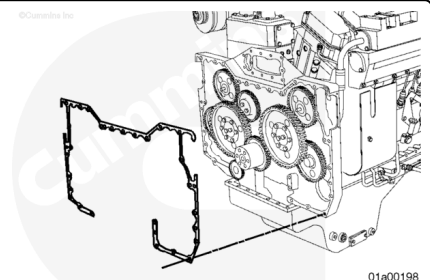
Note : Trim the gasket to the correct width since the oil pan is already installed.

Install the service gasket piece on top of the oil pan. Use a spray adhesive to hold it in position. Use Cummins sealant, Part Number 3823494, at the joints.



Use spray adhesive to help hold the gasket on the gear case during installation.

Install the gasket.



⚠ WARNING ⚠

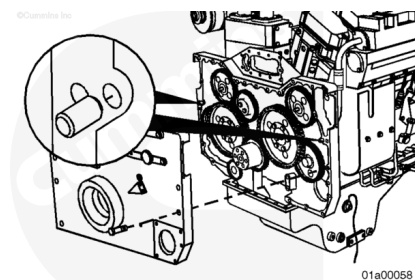
This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

Check that the two dowel pins are installed in the gear case.

Install the two guide studs to aid in the installation.

Use a hoist. Move the gear cover into position.

The use of a rubber mallet can be required to drive the gear cover against the gasket.

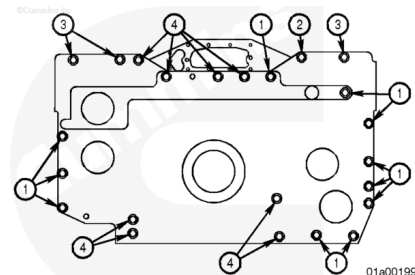


Install the capscrews into the front gear cover as shown.

1. M12 x 75 mm (11)
2. M12 x 145 mm (1)
3. M12 x 90 mm (3)
4. M12 x 130 mm (8).

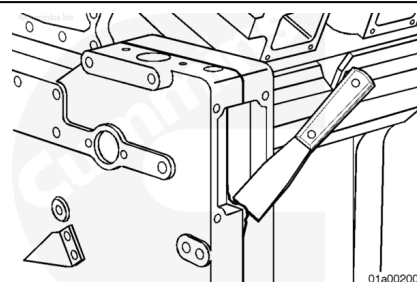
Tighten the capscrews.

Torque Value: 108 n•m [80 ft-lb]



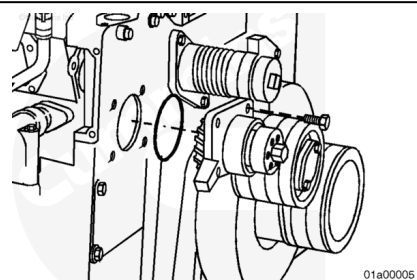
Trim any places where the gasket is protruding.

There are four places on the bottom of the front gear cover and front gear cover plate that will need the gasket trimmed.

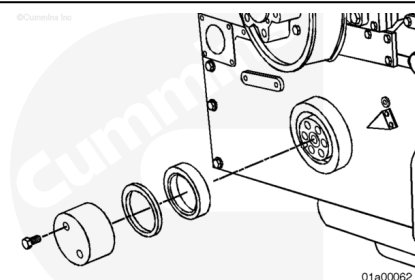


Install the charging system alternator drive. Refer to Procedure 009-008 in Section 9.

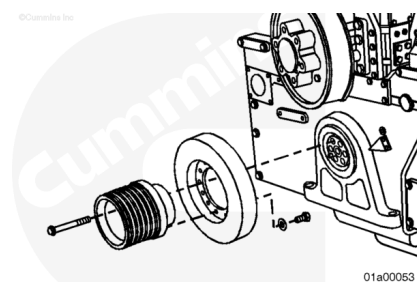
(/qs3/pubsys2/xml/en/procedures/57/57-009-008.html)



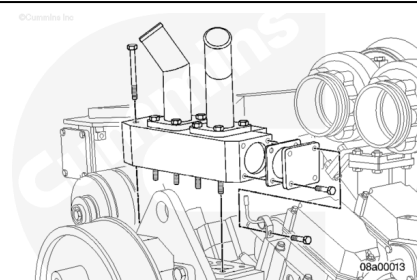
Install the front crankshaft seal. Refer to Procedure 001-023 (/qs3/pubsys2/xml/en/procedures/57/57-001-023-tr.html) in Section 1.



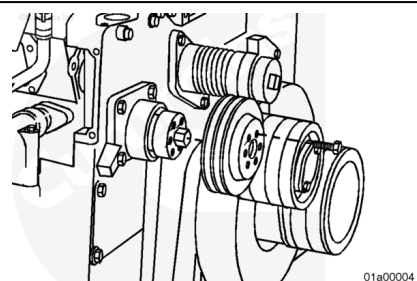
Install the vibration damper. Refer to Procedure 001-052 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-052.html>).



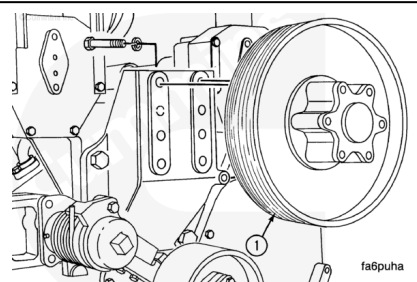
Install the thermostat housing and support. Refer to Procedure 008-015 (</qs3/pubsys2/xml/en/procedures/57/57-008-015.html>) in Section 8.



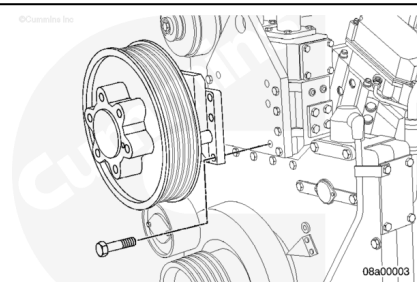
Install the charging system alternator drive pulley. Refer to Procedure 009-010 in Section 9. (</qs3/pubsys2/xml/en/procedures/57/57-009-010.html>)



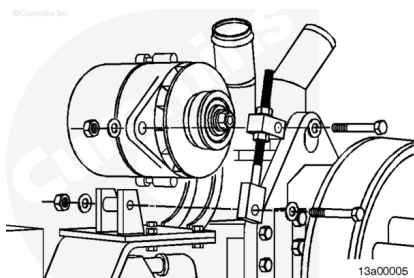
Install the fan hub idler assembly.



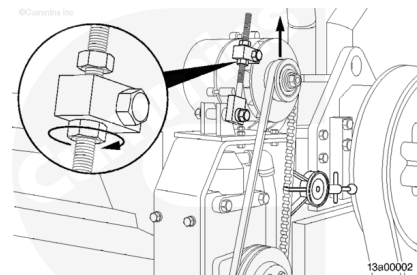
Install the fan hub. Refer to Procedure 008-036 (</qs3/pubsys2/xml/en/procedures/57/57-008-036-tr.html>) in Section 8.



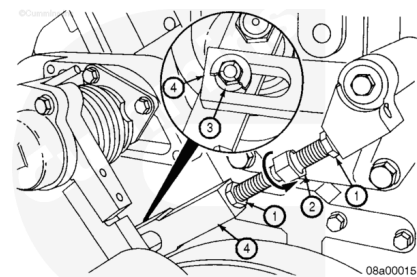
Install the alternator. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/57/57-013-001-tr.html>) in Section 13.



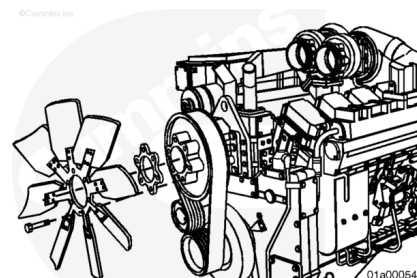
Install the belt. Refer to Procedure 013-005 in Section 13 (</qs3/pubsys2/xml/en/procedures/57/57-013-005-tr.html>).



Install the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/57/57-008-002-tr.html>) in Section 8.

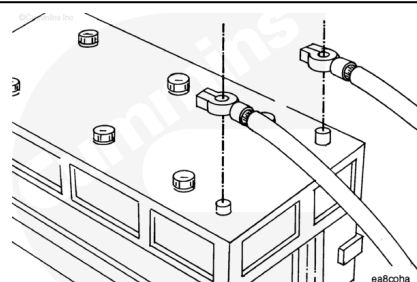


Install the fan spacer(s) and fan. Refer to Procedure 008-040 in Section 8 (</qs3/pubsys2/xml/en/procedures/57/57-008-040-tr.html>).



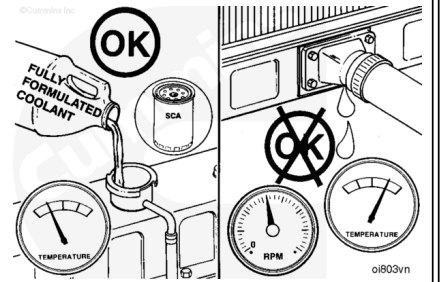
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Connect the batteries and air supply line to the air starter.

Fill the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/57/57-008-018-tr.html).



Last Modified: 19-Dec-2025
